

PyRestore

A Configurable Data Processing Framework for Street-View Perception Analysis

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<https://github.com/GuojialeGeographer/2026GProject>

From physical pixels to perceived places.

Two readings of street-view imagery — physical composition captured by segmentation and perceived qualities read by surveys or by vision-language prompts — are not the same evidence.

What street-view imagery already answers

Physical composition

Green View Index, sky share, enclosure — pixel-level estimates from segmentation such as Cityscapes fine-tuned SegFormer. Expressive of *what* the camera sees.

What it does not directly answer

Perceived qualities

Restorativeness, perceived safety, beauty, renewal need — subjective impressions a survey or a vision-language prompt can probe. Expressive of *how* observers feel.

What perception studies need beyond the pixels

Geographic coverage

Why a streetscape inventory fails if it ignores spatial distribution.

Comparable human references

Survey-derived references for calibration, paired to image records.

Calibration of scale

Recovery of the human score scale, not only rank ordering of images.

Each new perception study rewrites the same pipeline.

PRS, six-dimension survey, paired renewal — each iterates over steps that are not specific to it. PyRestore collects those steps and lets the study-specific content live in one task file.

What every new study repeats

- Image loading · model invocation
- Output-field organisation · coordinate joining
- Failure handling · spatial export
- Validation · GIS-ready packaging

Almost identical between studies — and re-implemented each time.

What PyRestore proposes

Shared operations stay in the framework.

Study-specific content lives in one task file — the perception question, prompt, output fields and optional human reference.

01 · Configurable pipeline

A reusable pipeline whose study-specific content lives in one task YAML — no rewrites for new questions.

02 · One spatial dataset

Joins CV-derived physical indicators with VLM-derived perception indicators into one analysis-ready table.

03 · Evidence-based evaluation

465 Shenzhen records, four task definitions, 75 automated tests passing with FDR-controlled spatial reads.

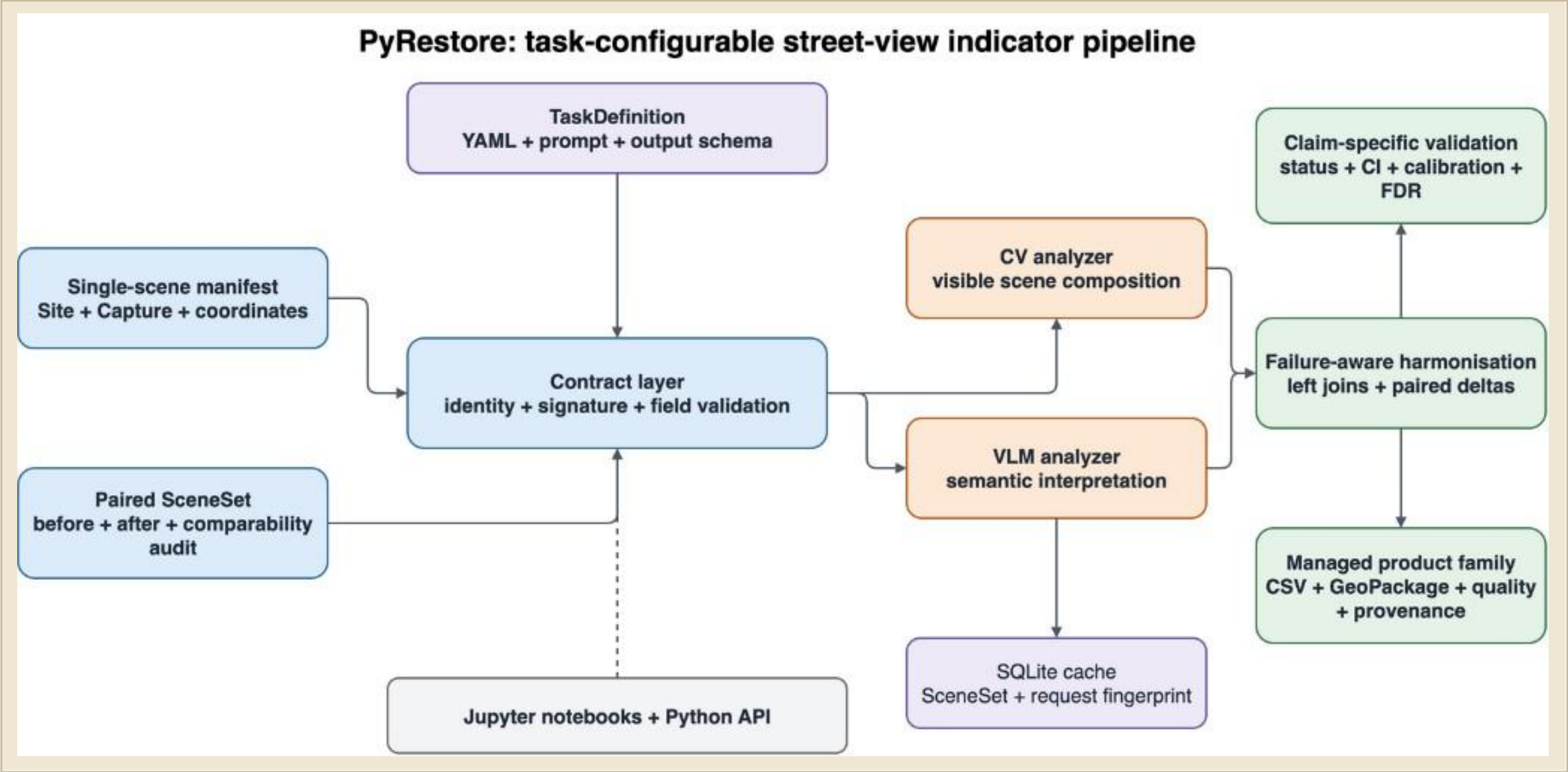
PyRestore sits inside a wider street-view perception literature.

Reference. Closely related work differs in input signature, configurability and how evidence is published. PyRestore keeps what is reusable and treats the study-specific content as a declaration.

<i>Work</i>	<i>Primary task</i>	<i>Configurable semantics</i>	<i>Input signature</i>	<i>Validation & GIS output</i>
Ma & Kwan [6]	Restorative quality of urban spaces	PRS-11 specific prompt · retrieval-augmented prior knowledge	Single scene	Human comparison · Shenzhen analysis
Perez & Fusco [7]	Prompt-defined streetscape assessment	Prompt configurable per task	Single scene	Point & street-segment mapping · task-level evaluation
Kim <i>et al.</i> [8]	Neighbourhood audit · human-centred agents	Domain-informed questions	Single scene	Structured annotations · human-centred assessment
Bai <i>et al.</i> [9]	Urban-renewal interpretation	Expert taxonomy · structured intervention types	Temporal pair	Validation subset · city-scale analysis
<i>PyRestore</i>	Configurable street-view analysis	Question · prompt · fields · reference all configurable	Single · audited pair	Quality checks · validation · FDR screening · GeoPackage

Comparable street-view perception workflows — PyRestore differs in treating shared operations as code and study-specific content as a YAML declaration.

One workflow, from standardised inputs to analysis-ready spatial datasets.



How to read it

Three readings.

Contract. Identity, signature, field validation.

Analyzers. CV and VLM run in parallel with a hash-keyed cache.

Products. CSV and GeoPackage staged behind run_manifest.json.

Figure 1. A manifest and a task declaration converge; physical and perception indicators are extracted and organised into datasets for validation, GIS mapping and further analysis.

Process and request state read together —
where each kind of failure can land.

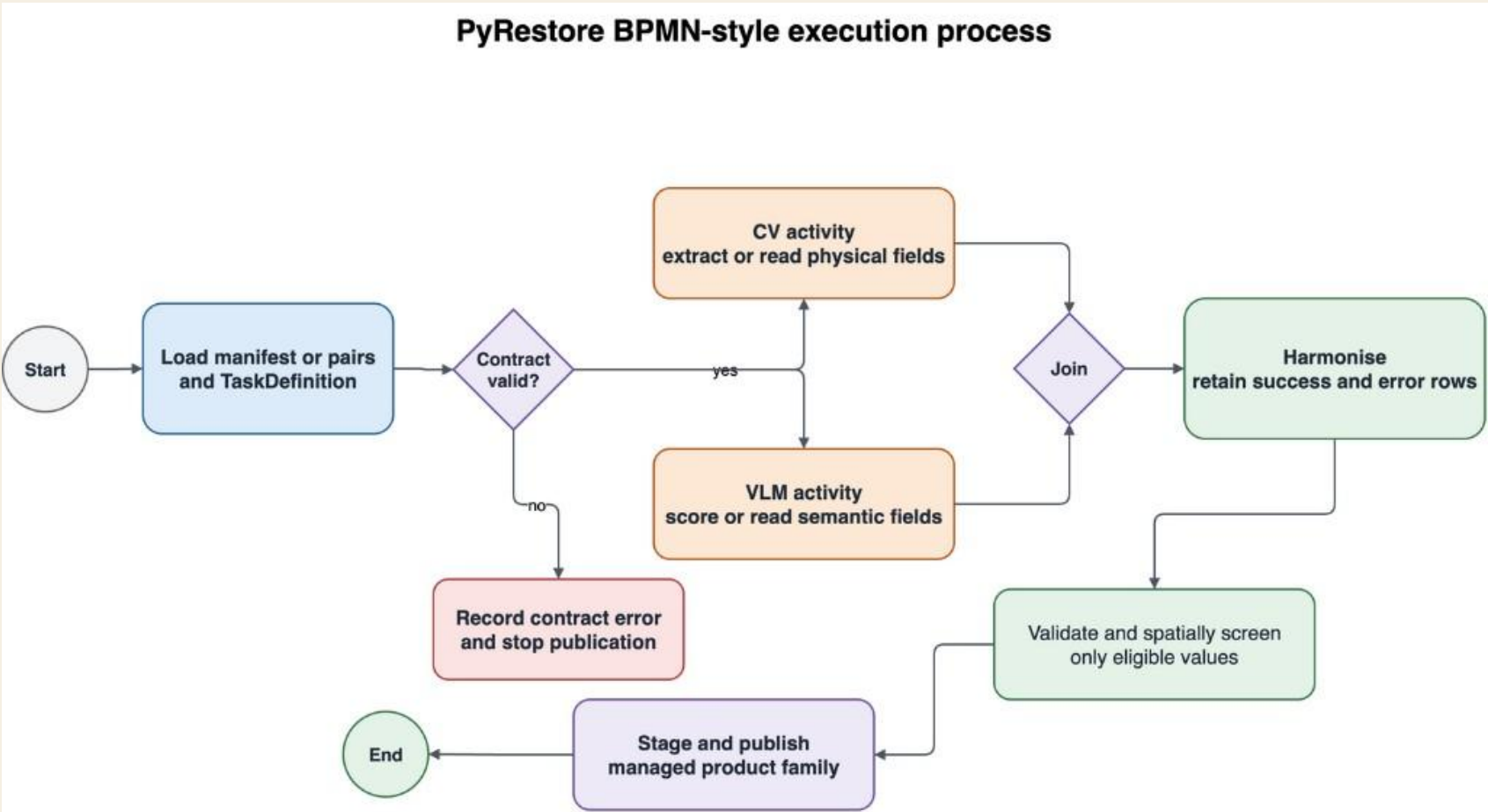


Figure 3 · BPMN-style execution

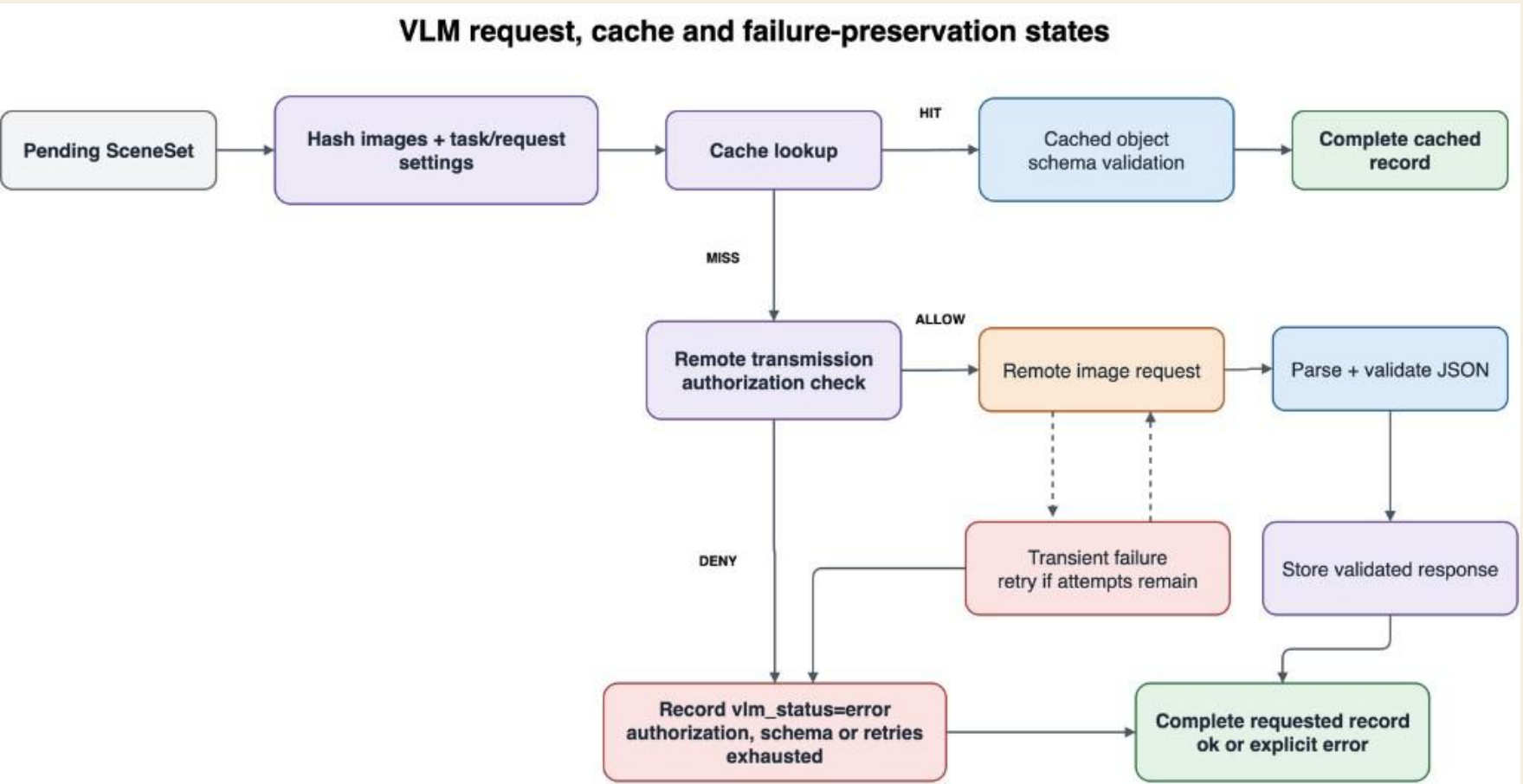


Figure 2 · VLM state machine

Three failure semantics — where the run stops

Contract. Stops publication. **Analyzer.** Stays in its row. **Spatial.** Conditional — eligible only.

Figure 3. Manifest or pair table meets a TaskDefinition; CV and VLM run in parallel before harmonisation, validation and staged publication.

Stable identity is the contract.

Task knowledge is a declaration, not a row.

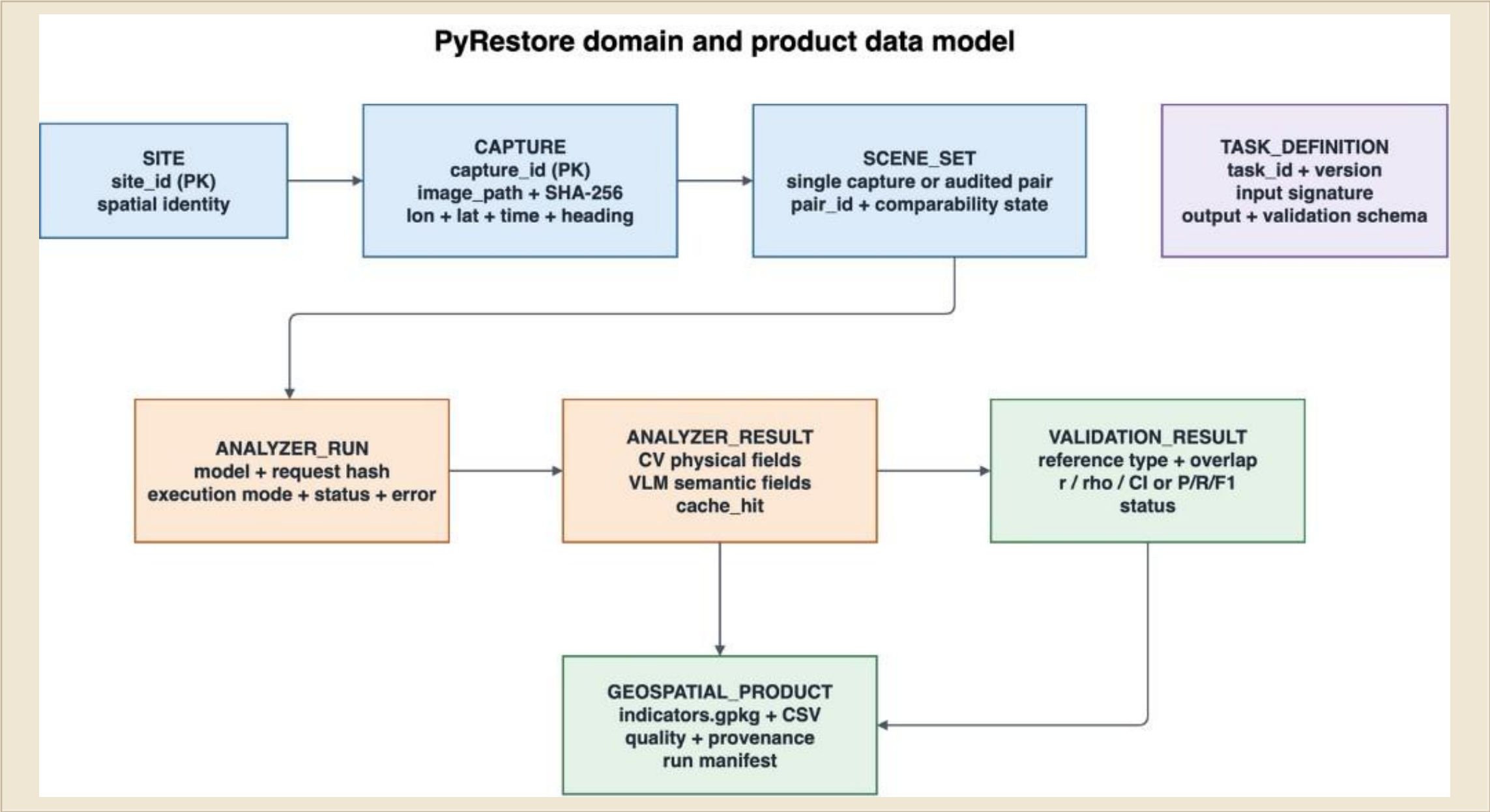


Figure 4. Site and Capture identities form SceneSets; analyzer runs feed optional validation; geospatial products package harmonised values.

Three things to read first

Identity, declaration, evidence.

Site · Capture. Spatial and observational identity are separated; image_path, content hash and metadata stay downstream.

TaskDefinition. A declaration object. It configures — it does not store evidence.

AnalyzerRun · Result. One operational attempt, one evidence family; source identity preserved.

One public API; clear boundaries between task semantics, runtime and execution.

Boundaries that matter

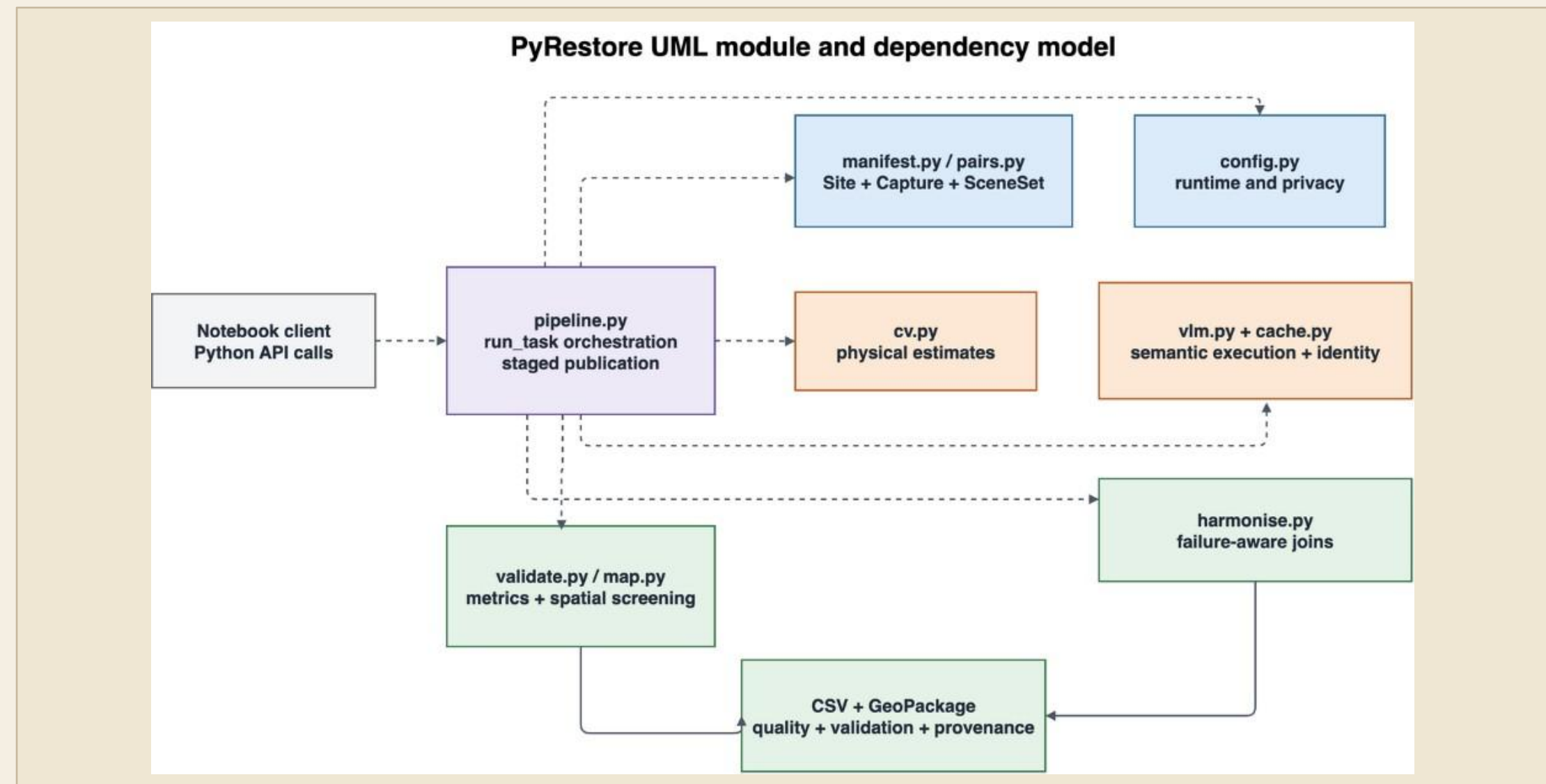
Three layers, one API.

Identity.manifest.py · pairs.py.

Runtime.config.py isolates credentials.

Orchestration.pipeline.py hosts *run_task*.

Evidence.cv.py, vlm.py + cache.py,
harmonise.py, validate.py, map.py.



Software evidence

Tests passing · 77.09% branch coverage · Ruff clean on package, tests, scripts and notebooks.

Calibration, ranking and spatial —
three readings of the 465-Shenzhen record.

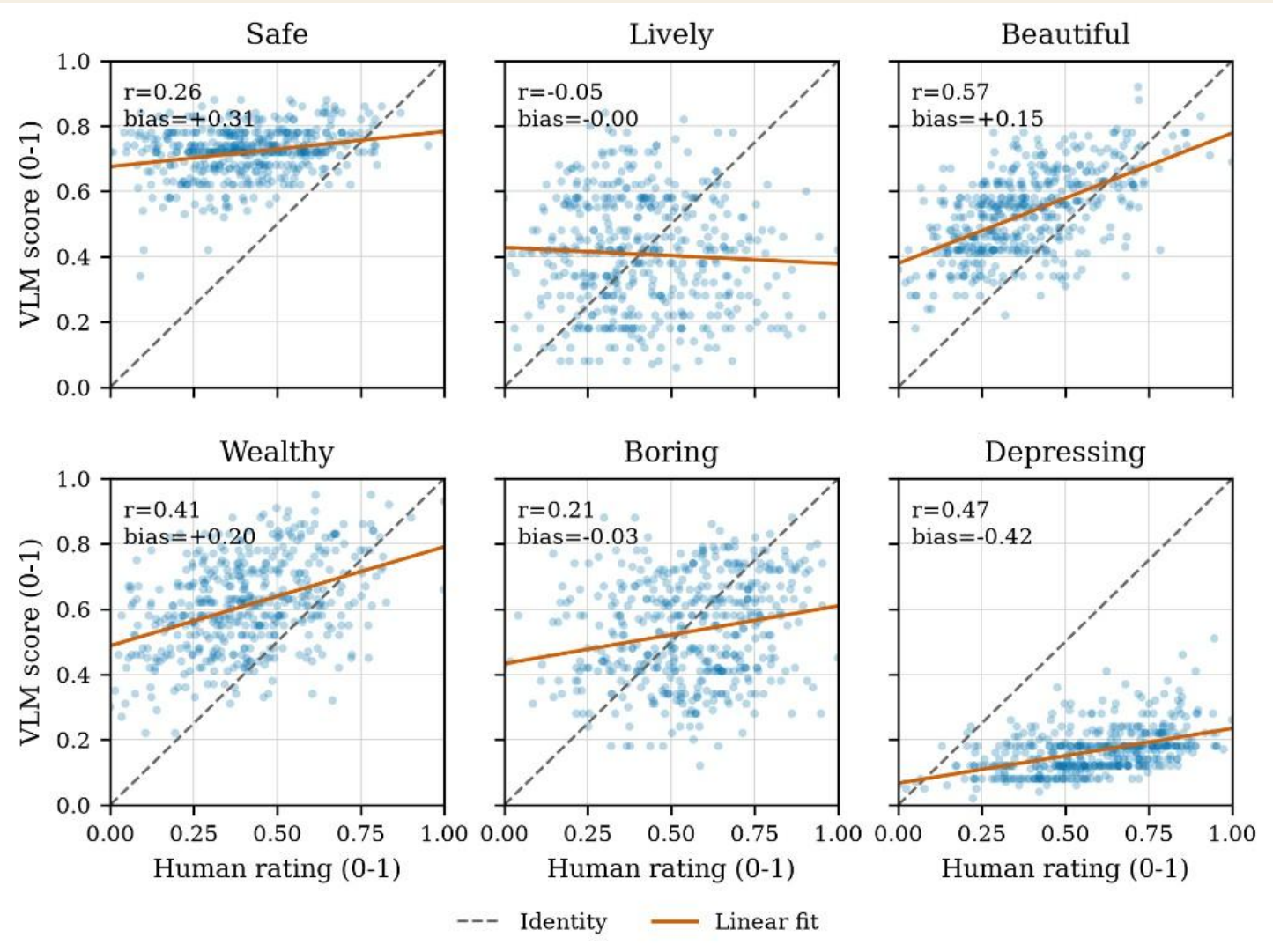


Figure 9 · six-dimension calibration

Field-level agreement · n = 465

Field	r	CI
Being-away	0.572	[0.51, 0.63]
Coherence	0.108	[0.02, 0.20]
Scope	0.372	[0.29, 0.45]
Fascination	0.162	[0.07, 0.25]
Four-dim mean	0.337	[0.25, 0.42]

What the case admits

Ranking varies from r = 0.108 to 0.572. Six-dimension predictions retain directional bias; never substitute the raw value for survey use.

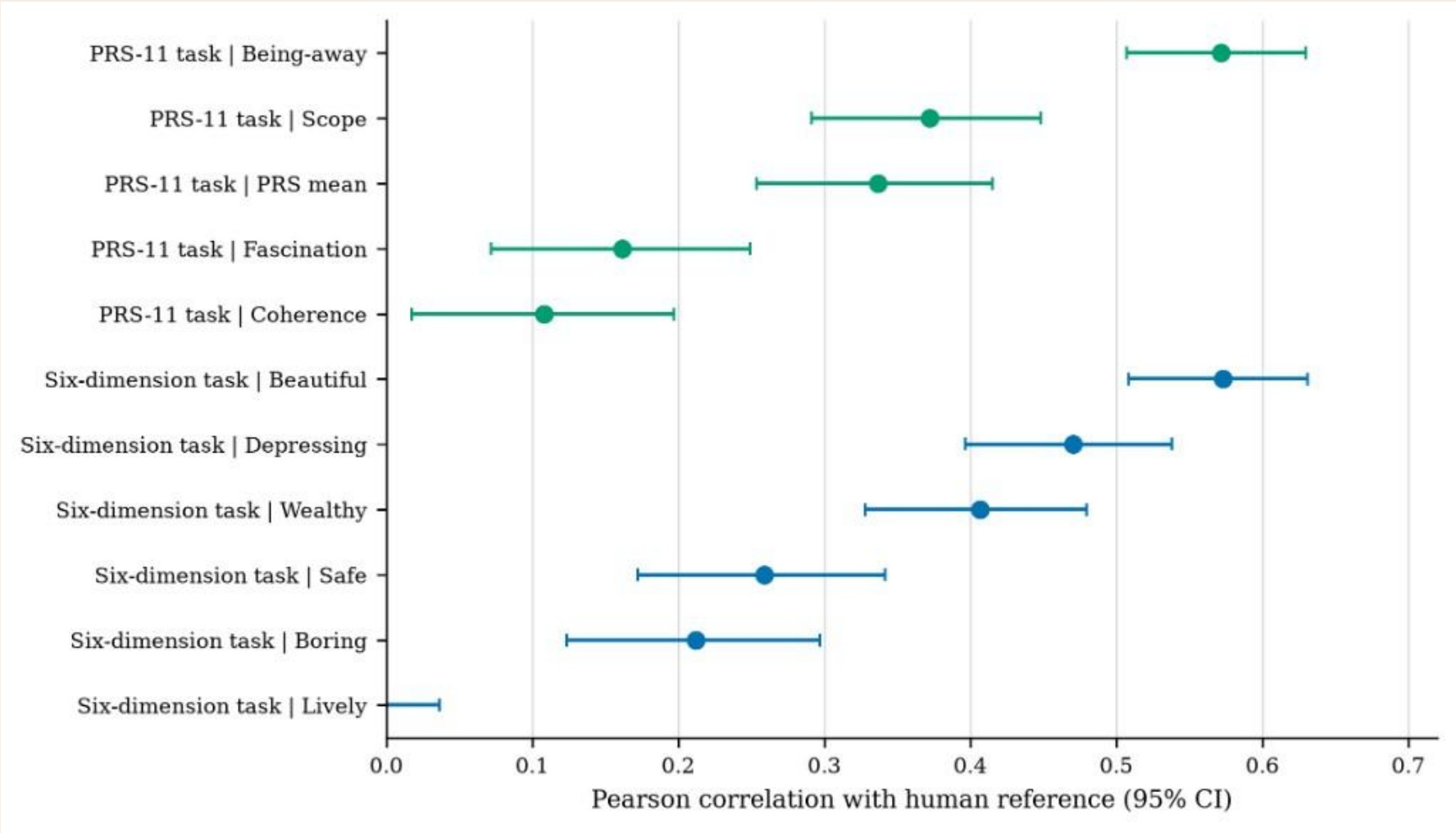


Figure 8 · cross-task agreement

Spatial · what survives BH-FDR

Global Moran I = 0.098 (p = 0.001); 0 of 40 unadjusted local candidates survive at k = 8 — the supported reading is global, not local.

The same city on the map;
three honest remarks, and an acknowledgement.

PRS-11 four-dimension mean (left), six-dimension Beautiful (right). EPSG:32649, fixed 0–1.

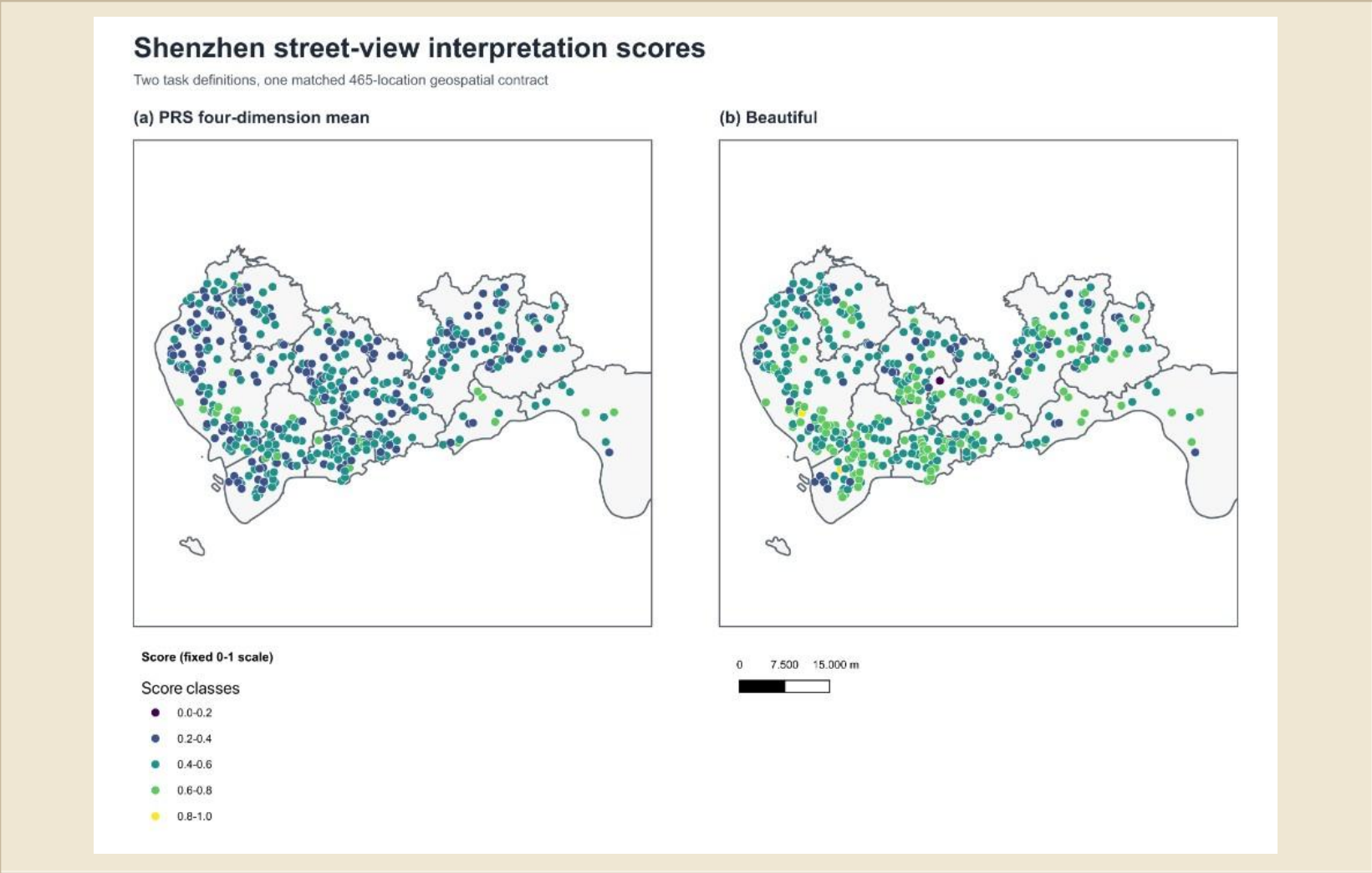


Figure 7 · QGIS view of 465 Shenzhen locations

With thanks to Prof. Daniele Oxoli and Prof. Lorenzo Gianquintieri for supervision, and to the Ma & Kwan Shenzhen data package.

Where the work stands

What PyRestore delivers

Reusable pipeline; one spatial dataset joining CV and VLM indicators; FAIR4RS-aligned outputs.

What remains unresolved

Field agreement varies; six-dimension predictions retain directional bias; the renewal task lacks a real temporal panel.

What follows next

Few-shot or retrieval-augmented prompts; a real before/after panel; sequence SceneSets.

Sustainable Development Goals

SDG 11 reproducible street-level evidence. SDG 9 reusable infrastructure aligned with FAIR4RS and OGC.